Dr. Ragadeepika Pucha

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Positions

- **Postdoctoral Researcher** **2023 – Present**
  Department of Physics & Astronomy, University of Utah

Education

- **Ph.D in Astronomy and Astrophysics** **2016 – 2023**
  Steward Observatory, University of Arizona
  Thesis: *Dwarf Galaxies As Probes of Galaxy Evolution*
  Advisors: Stéphanie Juneau, Arjun Dey
- **M.S in Astronomy and Astrophysics** **2016 – 2018**
  Steward Observatory, University of Arizona
  Advisors: Beth Willman, Jeffrey Carlin
- **Five Year Integrated M.Sc (Physics)** **2010 – 2015**
  Integrated Science Education and Research Center (ISERC)
  Visva-Bharati University, India

Selected Awards and Honors

- **2026: College of Science Outstanding Postdoctoral Researcher Award**, University of Utah.
- **2025: Outstanding Postdoctoral Researcher Award**, Department of Physics & Astronomy, University of Utah.
- **2022: FAMOUS grant**, American Astronomical Society.
- **2022: Best Poster Award**, Poster Symposium Targeting Early-career Researchers (PoSTER) Conference.
- **2020: Jamieson Graduate Fellowship**, University of Arizona.
- **2016 – 2017: College of Science Fellowship**, University of Arizona.
- **2015: International Max Planck Research School (IMPRS) Fellowship**, Max Planck Institute for Solar System Research, Göttingen, Germany.

Publications

My ADS Library: I have authored 20+ refereed journal papers, including 5 first-author papers and 2 significantly contributed papers. See my complete list of publications at the end.

Grants and Proposals

- NASA ADAP Proposal, Science PI
  Co-evolution of Dwarf Galaxies and Black Holes **2025 – 2027**
- Gemini Telescope Fast Turnaround Program 2024B, PI
  Discovery of the Lowest-Mass Black Hole in a Dwarf Galaxy **September 2024**
- DESI Secondary Target Program, PI
  Intermediate-Mass Black Holes in Low-Mass Galaxies **2021 – Present**

Technical Skills

- Programming: Python (primary), IDL, C, C++, SQL, GitHub, Slurm supercomputing.
- Softwares: ds9, GALFIT, GELATO, IRAF, Source Extractor, TOPCAT, LaTeX.
- Astronomy-related Python packages: astropy, astroconda, astrodrizzle, astroquery.
- `EmFit` - developed a Python-based spectral emission-line fitting code.

Scientific Collaborations

Dark Energy Spectroscopic Instrument (DESI)

- Galaxy & Quasar Physics AGN Topical Group Co-Lead **2024 – 2025**
- Member **2019 – Present**

Publicly Available Data Products

- `EmFit` Value-Added Catalog: Emission-line fitting results for low-redshift ($z \leq 0.45$) galaxies, *Pucha, R., et al. 2025, Pucha et al. 2026, submitted to AAS Journals*.
- Value-Added Catalog of Physical Properties of DESI Galaxies: Stellar masses and other physical properties from spectral energy distribution modeling, including AGN templates, *Siudek, M., Pucha, R. et al., 2024*.
- AGN/QSO Summary Value-Added Catalog: AGN and QSO Identifications for all targets in DESI, *DESI Galaxy and Quasar Physics Working Group including Pucha, R. et al., in preparation*.
- FastSpecFit Emission-line Catalog: Spectrophotometric fitting results from the `FastSpecFit` stellar continuum and emission-line modeling code, *Moustakas, J., including Pucha, R. et al., in preparation*.

Selected Talks and Posters

Invited Talks

- **2024:** DESI December Collaboration Meeting, Plenary Talk, *Tripling the Census of Dwarf AGN Candidates Using Early DESI Data*.
- **2022:** CosmoPalooza, 239th AAS Meeting, *Active Black Holes in Dwarf Galaxies with DESI Survey Validation*.
- **2021:** Plenary Talk, DESI Collaboration Meeting, *Active Galactic Nuclei in Low-mass Galaxies using DESI Survey Validation*.
- **2021:** ISM* Series, Space Telescope Institute (STScI), *Escape of Lyα in Lyman-Alpha Emitters: Dependence on Galaxy Properties and Environment*.
- **2021:** Astro Data Lab Special Session, 237th AAS Meeting, *Joint Spectroscopic and Photometric Analysis of Low-Redshift Galaxies*.

Contributed Talks

- **2026:** Empowering Science in the Data-Rich Era of Astronomy, NSF NOIRLab, *Scaling Up the Census of Active Black Holes in Dwarf Galaxies with DESI*.
- **2025:** Massive Black Holes Across Cosmic Time, Kavli Institute for Cosmology, *A Tenfold Increase in the Census of Dwarf AGN Candidates Using DESI DR1 Data*.
- **2025:** Quo Vadis Galaxy Evolution?, University of Heidelberg, *Exploring AGN and SF Feedback in Dwarf Galaxies: A DESI Perspective*.
- **2025:** Galactic Frontiers II Conference, Dartmouth College, *Active Black Holes in Dwarf Galaxies: A DESI Perspective*.
- **2023:** DESI December Collaboration Meeting, *Multi-Component Emission-Line Fitting in Low-Redshift DESI Targets*.
- **2023:** Coordinating the Next Generation of Spectroscopic Processing and Analysis Tools, NSF NOIRLab, *Emission-Line Fitting Using Astropy Modeling*.
- **2022:** What Drives the Growth of Black Holes?, *Black Hole Seeds in Dwarf Galaxies: Early Results from DESI*.

Posters

- **2022:** Poster Symposium Targeting Early-career Researchers (PoSTER) Conference, *Lyman-Alpha Escape from Low-mass, Compact, High-Redshift Galaxies* (best poster award).
- **2022:** Poster Symposium Targeting Early-career Researchers (PoSTER) Conference, *Active Black Holes in Dwarf Galaxies from DESI*.
- **2021:** iPoster-plus, 237th AAS Meeting, *Lyman-Alpha Emitters at $z \sim 2.65$: Probing the Low-mass Galaxies in the High-Redshift Universe*.
- **2018:** 231st AAS Meeting, *Wide-Field Structure of Local Group Dwarf Irregular Galaxy IC 1613*.

Selected Media Coverage

- **2026:** Astronomy, Active Black Holes are More Common than We Thought
  quoting me on the significance of a recent census of AGN candidates.
- **2025:** NOIRLab Press Release, *DESI Uncovers 300 New Intermediate-Mass Black Holes Plus 2500 New Active Black Holes in Dwarf Galaxies*
  featuring my research on the discovery of AGN candidates in dwarf galaxies.
- **2025:** The University of Utah News, *'Vast discovery' of Black Holes in Dwarf Galaxies*
  featuring my research on the discovery of AGN candidates in dwarf galaxies.
- **2025:** Space, *Largest-ever discovery of 'missing link' black holes revealed by dark energy camera*
  featuring my research on the discovery of AGN candidates in dwarf galaxies.
- **2024:** ScienceNews, *A Cosmic Census Triples the Known Number of Black Holes in Dwarf Galaxies*
  featuring my research on the discovery of AGN candidates in dwarf galaxies.
- **2022:** Berkeley Lab News Center, *Dark Energy Spectroscopic Instrument (DESI) Creates Largest 3D Map of the Cosmos*
  quoting me regarding science that can be done with DESI.

Selected Community Activities

- **2025:** *Tripling the Census of Active Black Holes in Dwarf Galaxies*, Public Talk, Utah Astronomy Club.
- **2019 – Present:** *Project Bharati:* An initiative for developing scientific interest among high-school and undergraduate students in India.
- **2021:** *Dwarf Galaxies as Probes of the Universe*, ISERC I/O online talk series, Visva-Bharati University.
- **2020:** *Importance of Women Education in India*, online workshop, Girl Up Bombay.
- **2020:** *Women in STEM*, online talk series, Stem4Gils, New Delhi.

Selected Teaching and Mentoring Activities

- **2023 – Present:** *Mentor* to graduate students, DESI Mentorship Program.
- **2019 – Present:** *Mentor* to >30 high-school and undergraduate students in India, Project Bharati.
- **2017 – 2022:** *Graduate Teaching Assistant*, University of Arizona
  - ASTR 202: Life in the Universe (Spring 2022)
  - ASTR 250: Fundamentals of Astrophysics (Fall 2020)
  - ASTR 203: Stars (Fall 2019, Fall 2021)
- **2018:** Volunteer, *Astronomy Tutoring for Major and Minors Program* (ATOMM), University of Arizona.

Research Students

- David Ohlson, Graduate Student, University of Utah **2025 – Present**
  Co-advisor with Prof. Yao-Yuan Mao and Prof. Anil Seth
- David Fowles, Undergraduate Student, University of Utah **2024 – 2025**
  Co-advisor with Prof. Yao-Yuan Mao
- Swagatha Bera, Undergraduate Student, Visva-Bharati University **2020 – Present**

References

- Dr. Yao-Yuan Mao, Assistant Professor, University of Utah, *yymao@astro.utah.edu*
- Dr. Stéphanie Juneau, Associate Astronomer, NSF NOIRLab, *stephanie.juneau@noirlab.edu*
- Dr. Arjun Dey, Astronomer, NSF NOIRLab, *arjun.dey@noirlab.edu*

List of Publications

Major Contributions

- **Pucha, R.**, Juneau, S., Mezcua, M. et al., Submitted to AAS Journals.
  A New Record Census of Dwarf AGN and a Bimodal $M_{BH} - M_{\star}$ Scaling Relation with DESI DR1.
- Morales, V., Mezcua, M., **Pucha, R.** et al., in DESI Collaboration review.
  The Largest Sample of AGN Outflows in Dwarf Galaxies using DESI DR1.
- **Pucha, R.**, Juneau, S., Dey, A., et al., **2025**, ApJ, 982, 10.
  Tripling the Census of Dwarf AGN Candidates using DESI Early Data.
- Siudek, M., **Pucha, R.**, M., Mezcua, M. et al., **2024**, A&A, 691, A308.
  Value-Added Catalog of Physical Properties of more than 1.3 million galaxies from the DESI Survey.
- **Pucha, R.**, Reddy, N., Dey, A., et al., **2022**, AJ, 164, 159.
  Lyα Escape from Low-mass, Compact, High-Redshift Galaxies.
- **Pucha, R.**, Carlin, J., Willman, B., et al., **2019**, ApJ, 880, 104.
  Hyper Wide Field Imaging of the Local Group Dwarf Irregular Galaxy IC 1613: An Extended Component of Metal Poor Stars.
- **Pucha, R.**, Hiremath, K.M., Gurumath, S., **2016**, Journal of Astrophysics and Astronomy, 37, 3.
  Development of a Code to Analyze the Solar White-Light Images from the Kodaikanal Observatory: Detection of Sunspots, Computation of Heliographic Coordinates and Area.

Minor Contributions

- Jewell, S., Smethurst, R., Lintott, C., including **Pucha, R.**, et al., Submitted in MNRAS.
  Evidence of Secularly-driven Galaxy-SMBH Co-evolution in 2,435 DESI-disk Galaxies.
- Alcolea, J., Siudek, M., Eriksen, M., including **Pucha, R.**, et al., in DESI Collaboration Review.
  Beyond Traditional Emission-line Diagnostics: Using Autoencoders to uncover Active Galactic Nuclei in DESI Spectra.
- Alfarsy, R., Canning, R., Mueller, E., including Pucha, R., et al., in DESI Collaboration Review.
  CIV Line Shape Analysis and Black Hole Mass Estimates of DESI Quasars.
- Alfarsy, R., Canning, R., Mueller, E., including Pucha, R., et al., Accepted for Publication in MNRAS.
  Stacked Reverberation Mapping of High Redshift Quasars in DESI. I. Feasibility Analysis.
- Mezcua, M., Laloux, B., Scialpi, M., including Pucha, R., et al., Accepted for Publication in A&A.
  Euclid: Quick Data Release (Q1) - Dual AGN in low-mass galaxies
- DESI Collaboration, including **Pucha, R.**, et al., **2026**, AJ, 171, 285.
  Data Release 1 of the Dark Energy Spectroscopic Instrument.
- Juneau, S., Jacques, A., Pothier, S., including **Pucha, R.**, et al., **2025**, ASPC, 541, 77.
  SPARCL: SPectra Analysis and Retrievable Catalog Lab.
- de los Reyes, M., Asali, Y., Wechsler, R., including **Pucha, R.**, et al., **2025**, ApJ, 989, 91.
  Stellar Mass Calibrations for Local Low-Mass Galaxies.
- Juneau, S., Canning, R., Alexandar, D., **Pucha, R.**, et al., **2025**, AJ, 169, 157.
  Identifying Missing Quasars from the DESI Bright Galaxy Survey.
- DESI Collaboration, including **Pucha, R.**, et al., **2024**, AJ, 168, 58.
  The Early Data Release of the Dark Energy Spectroscopic Instrument.
- DESI Collaboration, including **Pucha, R.**, et al., **2024**, AJ, 167, 62.
  Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument.
- Fawcett, V., Alexander, D., Brodzeller, A, including **Pucha, R.**, et al., **2023**, MNRAS, 525, 5575.
  A Striking Relationship between Dust Extinction and Radio Detection in DESI QSOs: Evidence for a Dusty Blow-out Phase in Red QSOs.
- DESI Collaboration, including **Pucha, R.**, et al., **2022**, AJ, 164, 207.
  Overview of the Instrument for the Dark Energy Spectroscopic Instrument.
- Hviding, R., Hainline, K., Rieke, M., including **Pucha, R.**, et al., **2022**, AJ, 163, 224.
  A New Infrared Criterion for Selecting Active Galactic Nuclei to Lower Luminosities.
- Nidever, D., Dey, A., Fasbender, K., including **Pucha, R.**, et al., **2021**, AJ, 161, 192.
  Second Data Release of the All-sky NOIRLab Source Catalog.
- Pat, F., Juneau, S., Böhm, V., **Pucha, R.**, et al., **2020**, ASP Conference Series, Vol. 525, 67.

Reconstucting and Classifying SDSS DR16 Galaxy Spectra with Machine-Learning and Dimensionality Reduction Algoriths.

- Carlin, J., Garling, C., Peter, A., including **Pucha, R.**, et al., **2019**, ApJ, 886, 109.
  Tidal Destruction in a Low-mass Galaxy Environment: The Discovery of Tidal tails around DDO 44.
- Schindler, J., Fan, X., McGreer, I., including **Pucha, R.**, et al., **2018**, ApJ, 863, 144.
  The Extremely Luminous Quasar Survey in the Sloan Digital Sky Survey Footprint. II. The North Galactic Cap Sample.

Contributed Data Tutorial Jupyter Notebooks

- *DESI Data Tutorials.*
- *Introduction to DESI First Data Release (DR1) at the Astro Data Lab.*
- *Introduction to DESI Early Data Release (EDR) at the Astro Data Lab.*
- *Comparison of Spectroscopy from Sloan Digital Sky Survey (SDSS) and Dark Energy Spectroscopic Instrument (DESI) Survey.*
- *Multi-wavelength Image Cutouts and SDSS Spectra of Active Galaxies with Extreme Emission-Line Ratios.*
- *Stacking SDSS Spectra of Galaxies Selected from the BPT Emission-Line Diagnostic Diagram.*